AR/VR ASSIGNMENT – 5

Marker-Based AR with Vuforia

1. Project Overview

Summary

This project demonstrates the implementation of a Marker-Based Augmented Reality application using Unity and Vuforia Engine. The application uses a custom-designed image marker to display a 3D globe model in real time. When the marker is viewed through a webcam or mobile camera, the globe appears anchored on top of the marker. The project showcases image target recognition, real-time object tracking, and AR visualization using Vuforia within the Unity environment.

Technical Approach

The project was developed using Unity and Vuforia Engine. Unity was used for designing the AR scene and integrating the 3D globe model. Vuforia Engine was used for image recognition and marker tracking functionality. A custom marker image was uploaded to the Vuforia Developer Portal and processed through the Target Manager to generate a tracking database. The generated database was imported into Unity. An AR Camera and Image Target were configured within the Unity scene, and the 3D globe model was parented to the Image Target so that it remains aligned with the marker during camera movement and tracking.

2. Step-by-Step Implementation

Vuforia Setup

1. Created an account in the Vuforia Developer Portal.
2. Generated a development license key for the project.
3. Created a device database using the Vuforia Target Manager.
4. Designed and uploaded a custom image marker.
5. Checked the feature rating of the marker for stable tracking.
6. Downloaded the generated database package for Unity.
7. Imported the Vuforia database package into the Unity project.

Unity Integration

1. Created a new Unity project for AR development.
2. Imported the Vuforia Engine package into Unity.
3. Enabled Vuforia support in the Unity project settings.
4. Added an AR Camera to the scene.
5. Added an Image Target object from the Vuforia Engine menu.
6. Configured the Image Target using the uploaded marker database.
7. Imported the 3D globe model into the Unity environment.
8. Added the globe model as a child of the Image Target.
9. Adjusted the scale and position of the globe model.
10. Tested the AR application using a webcam to detect the marker and display the globe.

3. Result

The application successfully detected the custom image marker and displayed the 3D globe model on top of the marker in real time. The globe remained properly aligned with the marker during movement, demonstrating stable image target tracking and successful AR augmentation using Vuforia Engine and Unity.

4. Conclusion

This assignment provided practical knowledge about Marker-Based Augmented Reality using Unity and Vuforia Engine. The project demonstrated how image targets can be used to place virtual 3D objects into a real-world environment. Through this implementation, concepts such as marker tracking, AR Camera configuration, database integration, and object augmentation were successfully understood and applied.